Name: **UIN:**

Graphene and 2D Materials: Science, Processing, and Applications (ChE 494)

Quiz 1: Science of 2D Materials

September 19, 2019 **40 Minutes** **Total: 100**

1. Graphene has an exceptionally well-formed cubic lattice structure.

   (A) True (B) False

2. Graphite-Carbon Nanotubes-Graphene-Fullerene (Arrange these materials with the following dimensions in the same order).

   (A) 3D-2D-0D-1D (B) 0D-1D-3D-2D (C) 2D-1D-0D-3D (D) 3D-1D-2D-0D

3. How many "surfaces" are there in a monolayer of 2D nanomaterial (For Example: Graphene)?

   (A) 1 (B) 2 (C) 3 (D) 6

4. Layered materials are solids with strong in-plane chemical bonds but weak out-of-plane, van der Waals bonds (anisotropic bonding).

   (A) True (B) False

5. 1D Confinement-2D Confinement-0D Confinement-3D Confinement (Confinement vs. Materials).

   (A) 3D-2D-0D-1D (B) 0D-1D-3D-2D (C) 2D-1D-3D-0D (D) 3D-1D-2D-0D

6. In Stone-Wales (SW) defect, four hexagons are transformed into two pentagons and two heptagons by rotating one of the C-C bonds by 60°.

   (A) True (B) False

7. How many number of atoms are in each hexagonal unit cell of a graphene lattice?

   (A) 6 (B) 3 (C) 2 (D) 12

8. Electrons in graphene obey linear dispersion relation and behave like massless relativistic particles.

   (A) True (B) False

9. Monolayer MoS_2 is an indirect bandgap semiconductor with bandgap (E_g) = 1.8 eV.

   (A) True (B) False

10. How much lattice mismatching is between graphene and white graphene?

   (A) 10% (B) 1.7% (C) 20% (D) 5%

11. In a layered crystal, the electronic wave function extends in three dimensions. However, after exfoliation electrons are constrained to adopt a 2D wave function, thus modifying the electronic band structure.

    (A) True (B) False

12. Monolayer Graphene, Monolayer MoS_2, Monolayer h-BN (Match these 2D materials with the following energy bandgaps in the same order).

    (A) 1.8 eV, 0.5 eV, 6 eV (B) 0 eV, 3 eV, 1.8 eV (C) 10 eV, 0 eV, 1.8 eV (D) 0 eV, 1.8 eV, 6 eV

13. Wide Bandgap Insulator + Atomically Smooth Surface + Crystalline (Match these properties with one of the following 2D materials).

    (A) MoS_2 (B) Boron Nitride (C) Graphene (D) WS_2

14. Which polytype of MoS_2 crystal structure represents: hexagonal symmetry, two layers per repeat unit, and trigonal prismatic coordination?

    (A) 1T (B) 3R (C) 2H (D) None

15. Boron nitride (BN) is a synthetic binary compound located in III and V group elements in the Periodic Table.

    (A) True (B) False

16. In graphene, the sigma (σ) bonds are delocalized electrons and pi (π) bonds are localized electrons.

    (A) True (B) False

17. In an indirect bandgap semiconductor, an electron can shift from the highest-energy state in the valence band (VB) to the lowest-energy state in the conduction band (CB) without a change in the crystal momentum.

    (A) True (B) False

18. Optical transparency of a single layer of graphene (monolayer graphene) is 100%.

    (A) True (B) False

19. Which 2D material can be used for deep ultraviolet (UV ≈ 6 eV) emission or detection?

    (A) Graphene (B) Boron Nitride (C) MoS_2 (D) WS_2

20. Hexagonal boron nitride (h-BN) is isostructural and isoelectronic to……………………………………………….

    (A) MoS_2 (B) Graphene (C) Carbon Nanotube (D) Fullerene (C_{60})